System Validation Report

Project: Student Attendance Management System. Activity: Part 3 testing of completed Sprint 1 and Sprint 2 work. Execution dates: 6 November 2026 (Part 3 run, branch `feature/testing-feedback`, base commit `438845cb25f6`) and 9 November 2026 (full regression on `main` after the timetable, semester classes, send-login and interface changes, BL-01 D-24 to D-28). No Sprint 3 was started.

Decision

The tested local functional, integration, browser and performance profiles pass. Full deployment and specification acceptance remains **open** because native platform/version/device checks and monthly availability are not established by this execution.

Evidence totals on 9 November 2026 are 552 automated passes with four skips; 15 passing cross-module integration methods included in that total; and 12 passing browser/viewport cases covering 36 role journeys and 240 checkpoints. All four declared performance criteria passed again on 9 November on the current revision. These are different measurement units and must not be added together as a single test count.

Plan alignment and traceability

The current Test Plan supplies 157 use-case cases; its four performance cases bring the original scope to 161. The existing documentation draft added 17 cases, this update adds another 17 for the current implementation, 12 more catalogue the coverage follow-up tests, and 17 cover the 9 November changes (FR-22 to FR-24), for a total of **224 catalogue cases**. 219 have passing mapped automated equivalents, one has direct browser evidence and four have passing local performance profiles. A mapped equivalent does not mean each literal manual script was separately entered.

The detailed scenarios, setup, parameters, steps, expectations, actual outcomes and method links are in `Test_Case_Document.md` . Earlier plan identifiers are preserved; the three earlier partial-evidence assertions have new explicit checks.

Functional and integration coverage

Plan area	Fresh evidence	Scope
UC-01 Login and role access	162 mapped-area methods passed	Partial platform/default scope
UC-02 to UC-05 and UC-14 Academic management, timetable and semester classes	116 mapped-area methods passed	Executed
UC-06 to UC-09 and UC-11 Attendance, corrections, reports, search and attendance advice	148 mapped-area methods passed	Executed
UC-10 Shortage alerts and notifications	32 mapped-area methods passed	Executed
UC-12 Dashboards, profiles, navigation and sessions	162 mapped-area methods passed	Partial platform/default scope
UC-13 Encrypted backup and atomic recovery	41 mapped-area methods passed	Partial platform/default scope
Cross-module workflow journeys	15 mapped-area methods passed	Executed

Area rows overlap, particularly accounts and dashboards; do not sum them. The exact case-to-method mappings are authoritative. The 15 integration methods exercise real password/TOTP onboarding, provisioning, marking/correction, consistent totals and exports, notification recovery, role and CSRF denials, encrypted recovery and rollback on

corrupt/inconsistent snapshots. Database concurrency tests execute on PostgreSQL. Email transport is captured locally.

Browser and responsive execution

Browser or engine	Version	Viewport	Result
chromium	156.0.8078.4	1440, 768, 360, 320 px	PASSED (9 Nov)
firefox	157.0	1440, 768, 360, 320 px	PASSED (9 Nov)
webkit	27.2	1440, 768, 360, 320 px	PASSED (9 Nov)
chrome	156.0.8078.80	1440, 360 px	PASSED (6 Nov, before the redesign)

Each case signs in as Admin, Faculty and Student with password/TOTP. Assertions cover required fields, tables, report empty states, incomplete attendance, explicit mark preservation, saved marks, corrections, student summaries and logout. Layout checks include standards mode, loaded local styling, control labels, readable table headings, page overflow and keyboard-accessible horizontal table scrolling. No uncaught JavaScript errors were recorded.

The narrow-screen cases emulate 768, 360 and 320 px viewports with touch. They are not physical Android/iOS tests. Playwright WebKit is not branded Safari. Current Edge, native Safari and the full latest-two-release matrix were not executed in this session. The installed Chrome version is recorded above; no claim is made that it is the latest release.

An intermediate browser attempt was rejected by the harness because the source fingerprint changed while the coverage dependency was being added. Its eight journeys passed, but that attempt is not accepted evidence. The final matrix reran all eight cases after the source was stable and passed its start/end fingerprint check. The original attempt records are retained in the supporting-files archive. The browser matrix was rerun on 9 November 2026 against the current interface (12 cases, all passed); branded Chrome was not repeated that day.

Performance and capacity

Case	Declared target	Observed result	Status
PT_01 Common requests	At least 95% within 2 s	1,500/1,500 within 2 s; p95 0.242 s	PASS
PT_02 Save 60 students	Each of three at most 3 s	Slowest 0.303 s	PASS
PT_03 Semester CSV	Each of three at most 10 s	Slowest 0.161 s	PASS
PT_04 Capacity and correctness	1,000 students; 750,000 marks; 50 users	1,800/1,800 requests successful; peak in flight 50	PASS

Results are from the 9 November 2026 run on revision 3999bbd (the plan, seed and revision checks matched at start and end); the 6 November run also passed (p95 0.196 s). The dated BL29-v1 plan was frozen before data creation. The baseline contains 1,000 students, 60 faculty, 17 sections, 85 assignments, 12,750 sessions and exactly 750,000 marks. Fifty distinct persisted authenticated sessions issue 30 synchronized common-request rounds, followed by three marking and three report repetitions with 49 background users. The measurement records 1,800 timed requests, separate from 50 warm-up requests.

Each 60-student save persisted correctly. The first crossing generated 60 in-app notifications and 60 captured emails; later repetitions generated no duplicates. The run added 180 marks. Each semester CSV contained 300 scoped student/subject rows representing 22,500 marks.

The benchmark ran after regression and browser execution had finished. It measures one Python process with 50 threads and loopback PostgreSQL, including middleware, rendering and commit callbacks. It excludes network/proxy delay, browser assets, password/OTP setup and external SMTP. Repeat on the intended deployment before accepting deployment latency. Current revision and plan/seed bindings match at the beginning and end of the run.

Defects found and fixed during testing

Five test cases failed on their first execution. Each defect was fixed, the case was run again and it passed. The Test Plan (section 14.3) records each fix with its cause and re-test.

Fix	Case	What failed	Fix and re-test
F-01	UT_12_2	Faculty dashboard counted a deactivated student, disagreeing with the below-threshold list (found in manual testing)	Count only active students (b62c8ba); regression test IT_12_5; PASS
F-02	UT_01_17	Test failed when a random password contained "&" (test defect)	Test compares escaped text (3999bbd); PASS in 6 repeats and the full run
F-03	ST_12_1	Admin dashboard scrolled sideways with a very long user name at 1440 px	Shorten long names in the top bar (b62c8ba); 12/12 browser cases PASS
F-04	ST_08_1	My attendance table headers squeezed at 768 and 320 px	Normal cells and minimum advice width (b62c8ba); 12/12 browser cases PASS
F-05	IT_01_6	SEC-07 required a self-service reset link (VAL-01, 6 Nov)	Requirement corrected (BL-01 D-23); PASS

Open validation items

ID	Finding	Status and next evidence
VAL-01	SEC-07 specified a single-use forgotten-password token expiring after 30 minutes. The implementation requests an Admin reset and emails a temporary password.	Resolved by specification change on 6 November 2026 (BL-01 D-23). SEC-07, FR-01 and UC-01 now describe the Admin-assisted reset, which IT_01_6-7 and UT_01_9-10 cover. No application change was made.
VAL-02	99% monthly institutional-hours availability requires a complete observed month.	Not measured. Collect scheduled deployment health observations during 08:00-18:00 IST Monday-Saturday; unknown slots are not passes.
VAL-03	Latest two browser releases, native Safari/Edge and physical Android/iOS coverage are incomplete.	Partial compatibility evidence. Execute the remaining named environments before full NFR-07 acceptance.
VAL-04	Three native Windows backup ACL/reparse checks skip on macOS.	Not rerun on current source. Run the suite on Windows; do not substitute historical results for this revision.
VAL-05	External mail delivery, deployed TLS/cookie settings and real scheduler operation are outside local test boundaries.	Deployment validation required; captured mail and command tests do not prove external transport or scheduling.
VAL-06	184 executable lines and 71 branch outcomes were not exercised in the 9 November instrumented run (95.10% line, 90.78% branch coverage).	Coverage gaps recorded; 126 of the lines and 31 of the outcomes are Windows-only backup storage paths. Run the suite on Windows and combine coverage to measure them. No coverage acceptance threshold was supplied by the attachment.

ID	Finding	Status and next evidence
VAL-07	The 9 November changes (dashboards, sidebar, timetable, attendance advice) altered pages measured by PT_01 to PT_04.	Resolved: PT_01 to PT_04 were rerun on 9 November 2026 on the current revision and passed. Repeat on the intended deployment before accepting deployment latency.

The default-configuration OTP test also skips because OTP_REQUIRED is explicitly set. Functional OTP behavior is exercised. Successful automated execution does not certify every security requirement independently.

Evidence and conclusion

Detailed logs, raw results, screenshots and supporting sources are retained in a separate local supporting-files archive; they are not included in this submission folder.

Fresh evidence: automated results, final browser matrix and screenshot references, declared performance plan, seed counts, and performance observations. Per-case outcomes are in case traceability. Source and retained-file hashes are in the execution manifest and SHA256SUMS.

The Part 3 testing reports are complete for the executions documented here. Release sign-off remains subject to the open validation items. No Sprint 3 was created; the 9 November rerun covers the changes recorded in BL-01 D-24 to D-28.